



SCOT FORGE PROJECT

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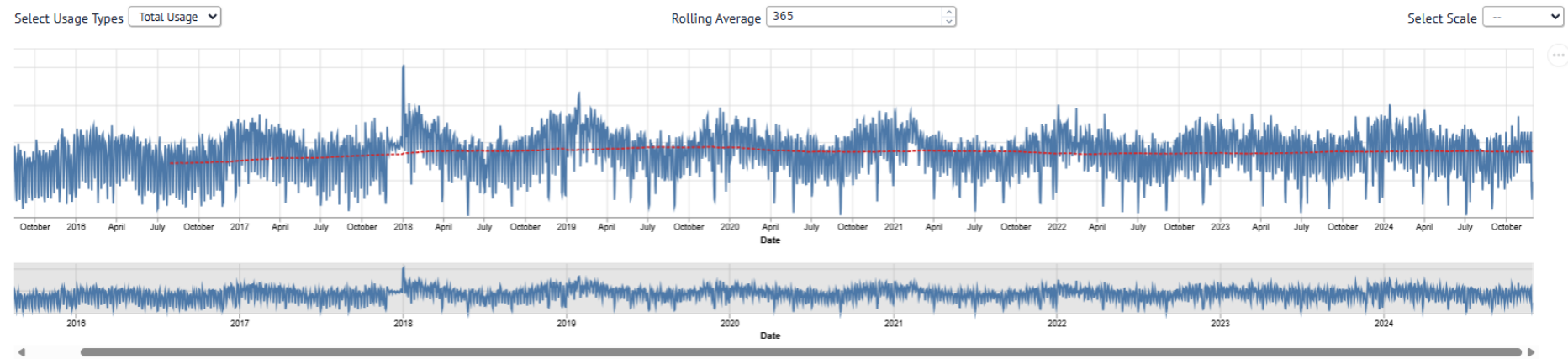
PROJECT OVERVIEW

- Prepare and analyse the provided natural-gas usage data and develop a time-series model that predicts next-day usage.
- Across the two plants, more than 60 furnaces may be active, but furnace-level gas usage is not currently measured. Instead, **Scot Forge** receives facility-level meter readings from a third-party provider.
- Recent Illinois legislation changed how natural gas tariffs are calculated, adding limits that can trigger penalties. These penalties depend on daily injections or withdrawals from the company's natural-gas storage bank and the total amount stored at the end of each month.

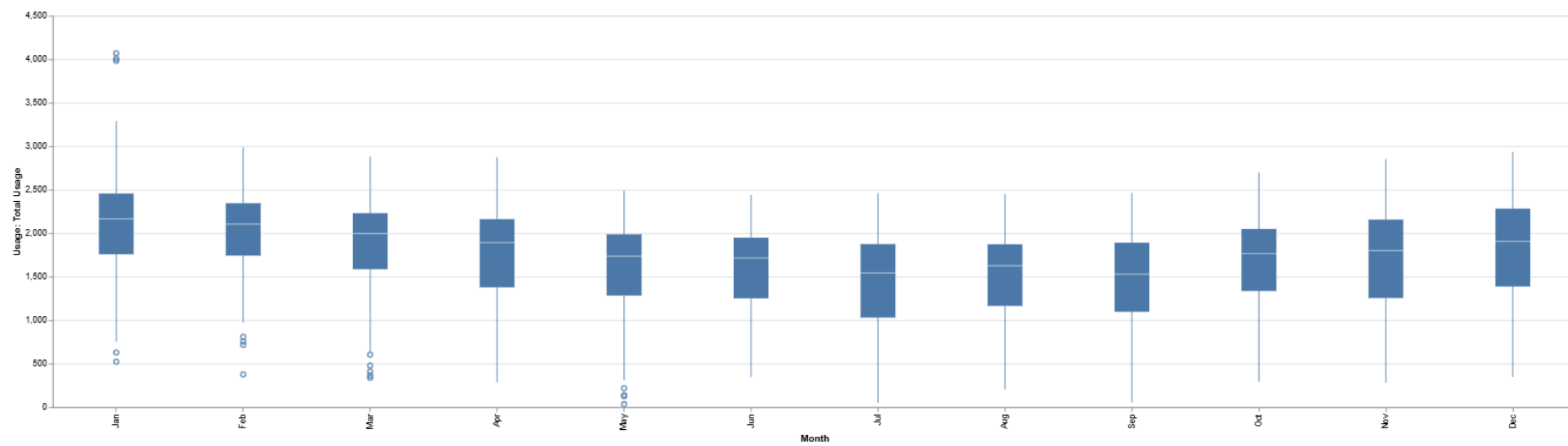
EXPLORATORY DATA ANALYSIS

GAS USAGE

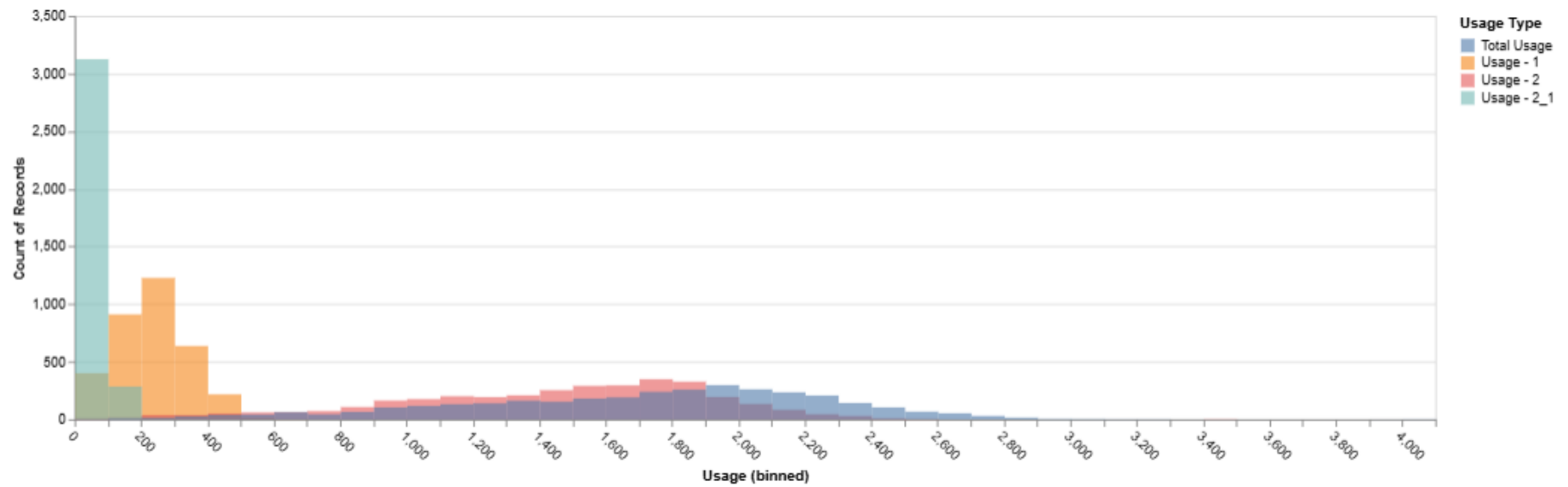
The lower panel provides an overview with a brush (selection) that controls the visible time window. The upper panel shows a detailed view of the selected time range. Use the dropdowns to choose which usage types and time scale to inspect. The y-axis is shared between panels to keep vertical scaling consistent when comparing ranges.



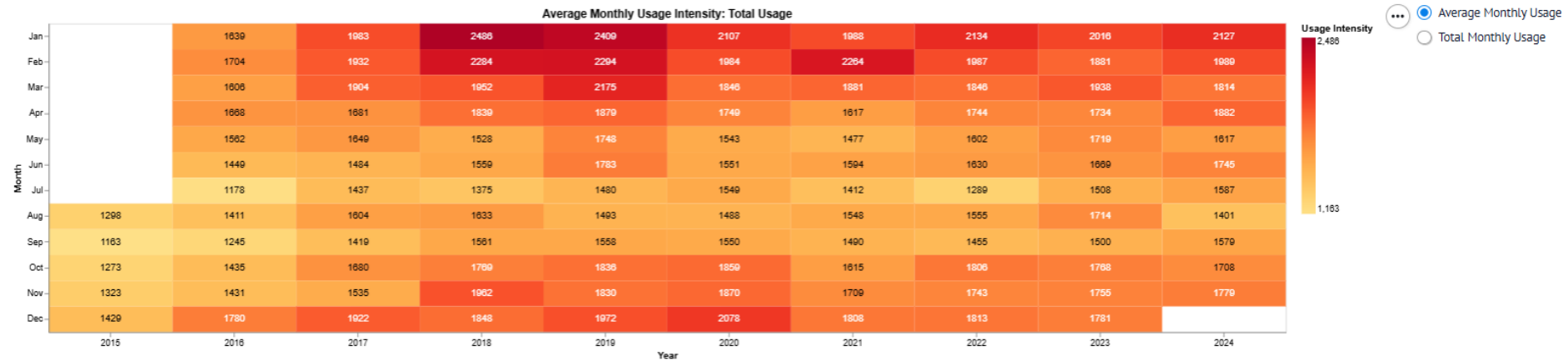
MONTHLY BOX PLOT



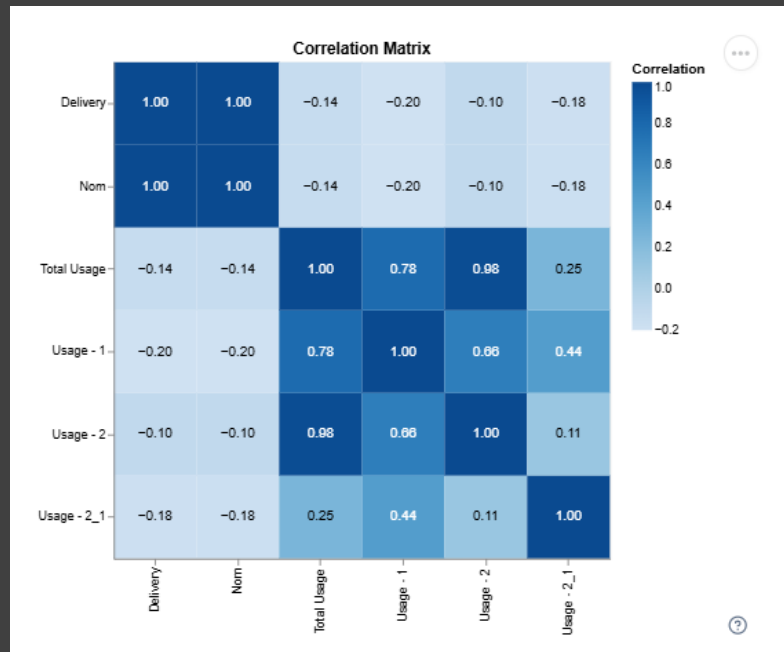
USAGE DISTRIBUTION



AVG MONTHLY USAGE INTENSITY



CORRELATION HEATMAPS



MODEL TRAINING & EVALUATION

WALK FORWARD

1

Train the model
using **all available
historical data
up-to-date (t-1)**

2

Generate a **2-day
forecast** everyday

3

Discard the
Forecast for **Day t**

4

keep the Forecast
for **Day t+1**

WALK FORWARD EXAMPLE

Data Till	Current date (t) and (t+1)						
	12/01/2024	12/02/2024	12/03/2024	12/04/2024	12/05/2024	12/06/2024	12/07/2024
11/30/2024	D	K					
12/01/2024		D	K				
12/02/2024			D	K			
12/03/2024				D	K		
12/04/2024					D	K	
12/05/2024						D	K

ASSUMPTIONS

- We allow for more wiggle room regarding small, insignificant errors that don't impact the bottom line.
- Large errors are not just "bigger", they are disproportionately more costly and damaging to our use case.
- Over-estimating and under-estimating carry the same penalty; we simply want to be as close to the target as possible.
- We care about the raw size of the error (e.g., missing by 10 units) rather than the error as a percentage of the total.

EXAMPLE

Assume the gas tank reserves are 2000 units and if we keep the units below 1900 and above 2100 we get penalise.

Case 1

- Predicted usage: 300 units and Actual usage: 100 units
- Absolute error = 200 units
- % Error = $200 / 300 = 66\%$

Case 2

- Predicted usage: 1500 units and Actual usage: 1300 units
- Absolute error = 200 units
- %Error = $200 / 1500 = 13\%$

In both case the absolute forecasting error is 200 units and will get penalise.

RMSE

- RMSE tells us how far off we are in actual gas units, while heavily penalizing large forecasting mistakes that create operational and financial risk.
- RMSE fits our assumptions very well as we want to penalise few large errors more than many small errors to avoid penalties

If errors are 5, 10 and 100 RMSE will be ~58 and MAE will be ~38. So RMSE penalise large error more which is what we want.

ALL MODELS

Model	Usage Type	RMSE
Basic Catboost	Usage - 1	37.93
Basic LGBM	Usage - 1	40.04
Catboost + Regime	Usage - 1	37.06
Ensemble	Usage - 1	41.70
LGBM + Regime	Usage - 1	36.72
LGBM + Temp	Usage - 1	40.31
Timesnet	Usage - 1	57.64
Basic Catboost	Usage - 2	239.56
Basic LGBM	Usage - 2	243.98
Catboost + Regime	Usage - 2	237.50
Ensemble	Usage - 2	227.66
LGBM + Regime	Usage - 2	252.51
LGBM + Temp	Usage - 2	249.19
Timesnet	Usage - 2	258.29
Basic Catboost	Usage - 2_1	8.88
Basic LGBM	Usage - 2_1	10.24
Catboost + Regime	Usage - 2_1	8.80
Ensemble	Usage - 2_1	8.17
LGBM + Regime	Usage - 2_1	7.84
LGBM + Temp	Usage - 2_1	9.87
Timesnet	Usage - 2_1	15.46

THANK YOU